Faster, Higher, Further Apart

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Abstract

In light of recent failures by the German Athletics Association (DLV) at the international level, this report examines the structural development of the extended national elite. Using a dataset of over 226,000 entries from the German rankings from 2001 to 2025, we analyze the impact of the COVID-19 pandemic and long-term trends in performance spreads. We utilize World Athletics Scores to make performances comparable across disciplines. Our results show a significant decline in performance during the pandemic (-2.57% median) and an increasing polarization of the national elite. While the absolute top performers improved, the extended elite declined. This decrease in performance density is particularly evident among men and in technical disciplines and suggests structural deficiencies in coaching presence and internal competition.

1. Introduction

As the world's largest athletics association, the Deutscher Leichtathletik-Verband (DLV) is tasked with facilitating peak individual performance and maintaining an internationally competitive national team (Deutscher Leichtathletik-Verband, n.d.). However, zero medals at the 2023 World Athletics Championships in Budapest and a historic low medal count at the 2024 Paris Olympics have left the organization in a sobering position. Although individual athletes still occasionally reach the podium, the declining return on investment in terms of Olympic success (Fremerey et al., 2024) raises questions about the underlying health of the national talent pool.

While medals are the most visible metric of success, they reflect only the peak of elite performance and offer little insight into how performance is distributed within the elite level. To examine potential structural shifts [or longitudinal developments], we analyze the top 35 rankings from 2001 to 2025 based on two research questions. First, we exploratively assess the impact of the COVID-19 pandemic as a short-term disruption to the performance development. Second, motivated by observed performance drop-offs beyond the top positions, we assess whether the performance gap within the elite has widened over time. Such a trend may reflect reduced internal competitive intensity, which can weaken international success, as elite development relies on strong competitive environments at the club level (Tihanyi, 2001, cited in Green & Houlihan, 2005).

Our analytical approach is detailed in Section 2, where we describe the data processing and justify the analytical framework. Building on this, we will analyze performance metrics to identify structural trends and evaluate our research questions (Section 3). Finally, Section 4 discusses the implications of these findings.

2. Data and Methods

2.1. Data Acquisition and Standardization

The data were retrieved from the official archive of the DLV, documenting top performances by German athletes in national and international competitions across major athletic disciplines (running, hurdles and steeplechase, jumping, throwing, and combined events) from 2001 to 2025. Performances are classified by sex (male, female) and age group (under 14 to open class).

Data were extracted from various PDF files using regular expressions to handle formatting variations across years. Outliers and entry inconsistencies (e.g., irregular time formats) were corrected via external sources where possible. Minor data gaps from parsing errors were manually completed by referencing the original files. Non-recurrent distances (e.g., 7.5 km youth events) were excluded, resulting in a final dataset of 238,187 entries.

We standardized discipline names, birth years and converted performance marks into numeric values (seconds for track events and meters for field events) to allow statistical analyses. To enable comparisons across disciplines, all performance marks were transformed into World Athletics scor-

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ing points (IAAF scores), which map heterogeneous results onto a common scale (Wor). As no official formula for score computation is publicly available, we implemented reverse-engineered, discipline-specific quadratic functions based on 2025 scoring tables (Chen, 2019). The following analyses are based on these standardized IAAF scores.

2.2. Data Selection and Grouping

To ensure a representative sample of elite performance, the analyses include the top 35 national results per year, sex, and age group. This threshold establishes a truncated distribution, focusing specifically on the high-performance tail of athletic performances. A ”Group” is defined as the combination of sex, age category, and discipline, consistent with official leaderboards. Although the dataset covers a broader range of age categories, the analyses focus on the U18, U20, and Adult groups, as these are the primary age groups competing at international championship level. To facilitate cross-disciplinary comparison, disciplines were aggregated into four primary clusters based on current team selection criteria (DLV): sprints, runs, throws, and jumps. Due to regulatory changes in hurdle heights and throwing weights, affected disciplines were excluded from median and rank-specific analyses but retained for gap evaluations to maintain longitudinal comparability.

2.3. Analytical Framework

Throughout this study, comparisons between time periods are reported as the relative percentage change. This value represents the percentage difference between the end and start periods, normalized to the start period. For performance levels, a positive value indicates improvement. For performance spread, a positive value indicates a widening gap. General performance shifts are assessed based on the relative change between the medians. These are selected for their robustness against outliers and the non-normal distribution of the data. Statistical significance is determined using the Mann-Whitney U test. To quantify the performance distribution, we analyze the relative change of a specific measure of performance variance. Significance is determined using bootstrap resampling (B = 10,000), which is optimal for our small sample size (N = 12). We calculated 95% confidence intervals (CIs) for the differences in performance variance. An CI containing zero indicates no significant change in performance density. To investigate the impact of the COVID-19 pandemic and to get an overall impression of the data, an exploratory analysis was conducted. A heatmap displaying the number of competitive events per month and year was used to examine seasonal competition patterns and identify disruptions in event frequency and data density during the pandemic period.

To isolate the effect of the pandemic year, performances were grouped into two cohorts: a pre-pandemic baseline comprising pooled IAAF scores from 2015–2019, and a pandemic cohort consisting of all IAAF scores recorded in 2020. General performance shifts were assessed using the relative percentage difference between period medians. Changes in performance density were evaluated using the interquartile range (IQR). The median and IQR were selected as robust measures of central tendency and dispersion, as our data are non-normally distributed and may include extreme values. ?The primary indicator of distributional change was the relative percentage difference in IQR between 2020 and the pooled baseline period. To analyze long-term structural changes, we compare the five-year periods 2001–2005 and 2021–2025. This timeframe smooths out annual fluctuations and, through a broader data base, ensures the statistical validity of the results. The analysis is conducted on two levels: 1. Recording performance trends and density for all groups to identify group differences. 2. Examining the development to identify the structure in which performance changes occur. This allows us to identify general trends and determine the structure of the changes.

To measure the performance density, we define the performance gap (G_p) between the three best and the three worst (e.g., 33rd–35th place) in our data set for the two time periods. The averaging of the top 3 reflects the international standard of success (podium), while the bottom 3 provide a stable and comparable baseline. We used the performance gap to measure the breadth of performance at the top level and utilized all available information (e.g., rankings) from our dataset to measure polarization. Mathematically, the gap (G_p) is defined as follows:

$$G_p = \frac{1}{3|P|} \sum_{y \in P} \left(\sum_{r=1}^{3} x_{y,r} - \sum_{r=33}^{35} x_{y,r} \right) \tag{1}$$

where $x_{y,r}$ represents the individual performance of rank r in year y, and $|P|$, in our case $|P| = 5$, represents the number of years per period. The development of the performance gap was calculated as the relaive percentage change between the two five-year periods.

To analyze the development of the different rankings, we calculate the arithmetic mean for each rank across all groups for the start and end periods. For each rank, even when differentiating by age group and gender, the sample size is at least n = 80. According to the central limit theorem, we can assume a normal distribution and thus calculate the arithmetic mean. The development of each rank was determined by calculating the percentage change between the start and end periods, normalized to the start period. A positive value indicates an improvement in the average ranking, a negative value a decline.

During the period under review, several rule changes were implemented regarding hurdle heights and throwing weights, particularly in the U18 (2012) and U20 age groups (2001) (Pro). Therefore, the affected disciplines were excluded from median and rank analyses, as changes in these parameters are directly attributable to equipment changes and not to athletic development. However, the performance gap analysis was retained, as this variance can be calculated independently of the absolute values.

3. Results

Analysis of the distribution of seasonal events reveals a significant temporal shift in the competition calendar. In 2020, events were concentrated in the latter part of the year, which marked a clear departure from the traditional midsummer peak. This shift toward late summer and autumn remains partially visible in 2021, though the schedule shows a gradual return to the pre-pandemic seasonal norm.

The Mann-Whitney-U test confirms a significant decline in performance levels across all athletic categories during the 2020 season compared to the 2015–2019 baseline ($p < 0.05$). The overall median IAAF score decreased by -2.14%, falling from 935 to 915 points. Throws saw the largest decline (-2.84%), followed by Jumps (-2.45%) and Sprints (-1.94%), while Runs remained most resilient (-0.85%).

The bootstrap analysis of the Interquartile Range (IQR) reveals significant shifts in performance dispersion during the 2020 season compared to the 2015–2019 baseline. Bootstrap analysis revealed significant increases in performance dispersion for Sprint ($+12.04\%$) and Run ($+6.41\%$), while the other disciplines Throw, Jump and remained statistically stable.

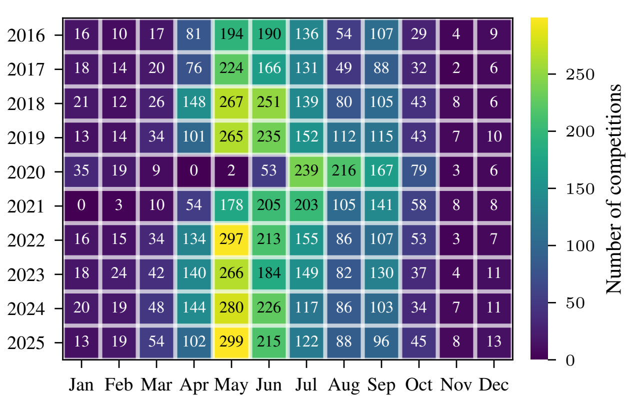

Figure 1.

Figure 3 illustrates a long-term performance shift across 106 groups between 2001-2005 and 2021-2025. While median trends were nearly balanced (38 improvements vs. 37

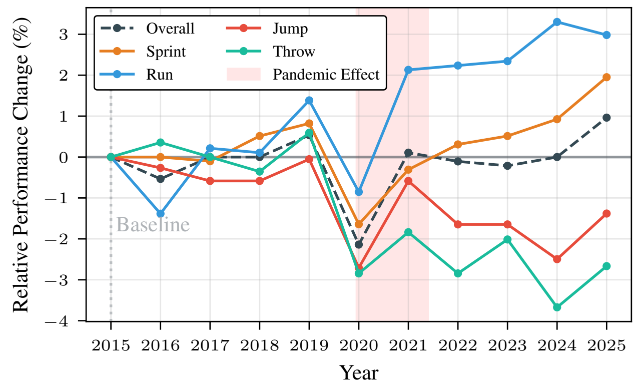

Figure 2.

declines), performance density decreased significantly: the performance gap widened in 49.1% of groups ($n = 52$) and narrowed in only 4.7% ($n = 6$). Running and sprinting were the top performers, with improvements occurring 2 to 3 times more often than declines. In contrast, technical events saw a major decline: throwing events were six times more likely to show a decrease in performance than an improvement, while jumping saw a 2.6 to 1 negative ratio.

Figure 4 illustrates a linear polarization of performance. While the top-ranked company improved by an average of 1.61%, the 35th-rank declined by 1.51%, with the turning point towards negative growth being reached at the 22nd-rank. This polarization is not apparent only among adult women; all ranks from third place onward improved more than the top two ranks. Overall, women in all age groups perform better than their male counterparts. The most significant declines were seen in the male age groups under 18 and under 20. In the under-18 age group, the decline begins as early as rank 6, while in the under-20 age group it starts as early as rank 9.

4. Discussion & Conclusion

The results demonstrate a systematic widening of the performance gap, particularly among men and young athletes. This increase in the performance range correlates directly with the 11.9% decline in German Athletics Association (DLV) membership and the massive 26.6% loss in the 27- to 60-year-old age group (Lei). This is the largest group of coaches, and their decline is particularly noticeable in technical disciplines, which rely on more intensive coaching.

Our analysis of the years during the pandemic also reveals a difference between disciplines. Running disciplines improved thanks to carbon technology (Willwacher et al., 2023) and uninterrupted training blocks (Venkatesan, 2022). Training facility closures led to a measurable drop in per-

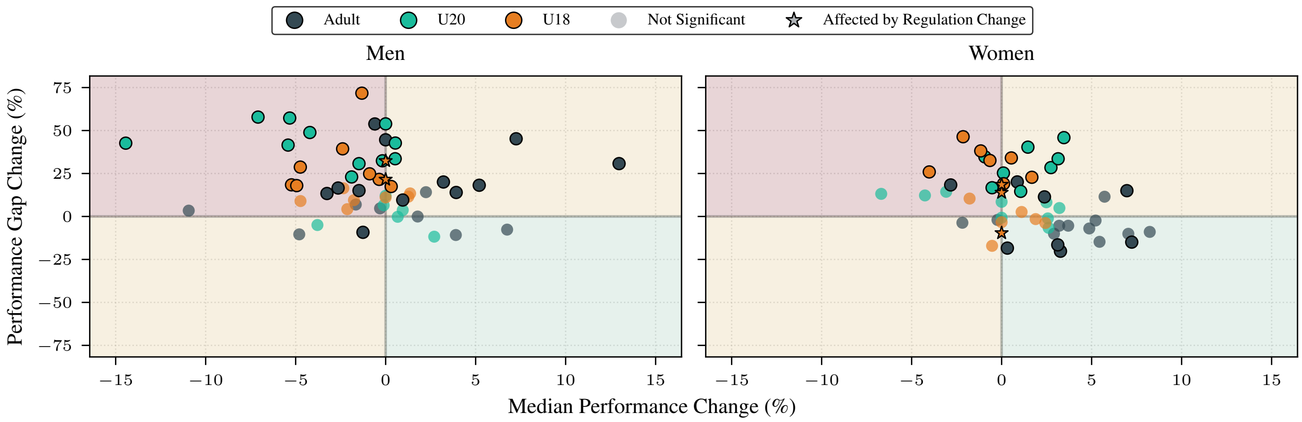

Figure 3. Change in median performance vs. performance gap (2001–2005 to 2021–2025, in %) for men and women across age groups. Marker transparency denotes statistical significance for performance gap (p ¡ 0.05). Median changes for disciplines with regulatory updates are fixed at zero due to restricted comparability.

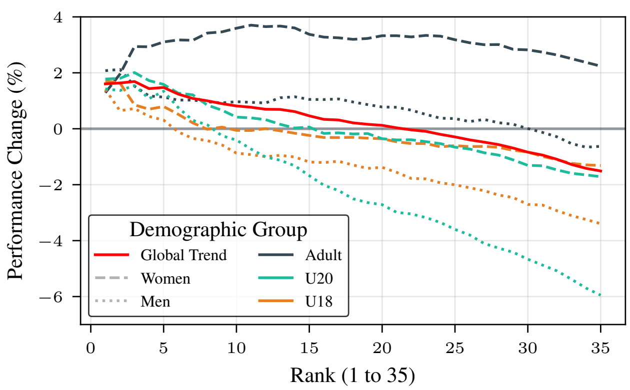

Figure 4. Average rank-specific performance development (2001–2005 to 2021-2025, in %), by age group and sex.

formance in the technical disciplines. Women, on the other hand, show a catching-up effect, suggesting that professionalization is not yet complete.

The increasing polarization can also be attributed to the distribution of resources. National team athletes are secured through centralized funding. The extended elite relies on external support, which was truncated during the COVID-19 pandemic (Breuer et al., 2021). As a result, this extended elite has been lost. This lack of internal competition weakens international competitiveness in the long term, according to the adage ”Competition breeds excellence.”

The analysis is limited to the top 35 and thus represents the extended elite, not grassroots sports. This should be analyzed further in the future to understand the overall structural changes. The trend towards polarization explains the lack of medals at international championships. Future focus must be on strengthening the extended elite and coach training to revitalize competition, which will also benefit the top athletes.

Contribution Statement

The authors confirm their contributions to this work as follows: Maximilian Wernz performed data parsing, visualization, and Git repository management. Mattis Dietrich conducted the performance gap analysis. Luca Bouché performed the analysis of the impact of the COVID-19 pandemic. Eric Müller performed manual data cleaning. All authors conducted exploratory data analysis and contributed equally to the report.

References

DLV-kaderbildungsrichtlinien und -normen 2026 veröffentlicht. URL https://www.leichtathletik.de/aktuelles/news/news-detail/80502-dlv-kaderbildungsrichtlinien-und-normen-2026-veroeffentlicht.

das leichtathletik-portal - downloads. URL https://www.leichtathletik.de/service/downloads.

Progression of world athletics records. URL https://media.aws.iaaf.org/misc/eBook/index.html#page=483.

World athletics technical information | official documents. URL https://worldathletics.org/about-iaaf/documents/technical-information.

Breuer, C., Dallmeyer, S., and Steinfeldt, H. Die ökonomischen auswirkungen von COVID-19 für nachwuchsleistungs- und spitzensportler:innen in deutschland. 2021.

Chen, J. Calculating World Athletics Scoring Coefficients. https://jeffchen.dev/posts/Calculating-World-Athletics-Coefficients/, 2019. Accessed: 2025-01-20.

Deutscher Leichtathletik-Verband. Wir im dlv. https://www.leichtathletik.de/wir-im-dlv, n.d. Accessed 6 Jan. 2026.

Fremerey, M., Gerards Iglesias, S., and Schlößer, S. Olympia: Mehr geld, weniger ertrag. IW-Kurzbericht 56/2024, Institut der deutschen Wirtschaft (IW), Köln, 2024. URL https://hdl.handle.net/10419/302126.

Green, M. and Houlihan, B. *Elite Sport Development: Policy Learning and Privilege*. Routledge, London, 2005. doi: 10.4324/9780203022245. URL https://doi.org/10.4324/9780203022245.

Venkatesan, S. Running performance during the COVID-19 pandemic. 8(4):229, 2022. ISSN 2469-5718. doi: 10.23937/2469-5718/1510229.

Willwacher, S., Mai, P., Helwig, J., Hipper, M., Utku, B., and Robbin, J. *Does advanced footwear technology improve track and road racing performance? An explorative analysis based on the 100 best yearly performances in the world between 2010 and 2022*. 2023. doi: 10.51224/SRXIV.297.